Informational Power and Institutional Design: When a Hegemon May Choose Consensus

Manoel Galdino (University of São Paulo; mgaldino@usp.br)

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Abstract

When and why would a hegemon prefer consensus to majority rule in an international organization? I argue that consensus can be a technology of hegemonic power rather than a constraint upon it. In a formal model, a hegemon privately observes the value of cooperation and uses Bayesian persuasion to influence weaker states' entry decisions. Under majority rule, weak states can bypass the hegemon, eliminating any screening problem. Under unanimity, weak states must include the hegemon in every bargaining coalition without knowing its type, creating a screening cutoff at which their behavior changes discretely. This generates an upward jump in the hegemon's value function—a non-convexity that Bayesian persuasion exploits. The institutional comparison has a single-crossing property: there is a unique prior threshold, in closed form, above which the hegemon prefers unanimity and below which majority can dominate by making entry easier. The mechanism reframes institutional design as a trade-off between informational leverage under unanimity and easier participation under majority, providing a distributive explanation for consensus rules in organizations like the GATT/WTO.

1 Introduction

Most international organizations operate by consensus, defined as decisions requiring the approval of all or nearly all participants, even when powerful states drive their creation (Gould 2022). Why would a powerful state ever prefer consensus over majority rule, where decisions can pass with only a simple majority? This is a puzzle, as the international arena is where material asymmetries—that is, differences in resources and influence among states—should matter most. If institutional rules reflect power, powerful states might be expected to favor arrangements that let them turn that power into a distributive advantage, meaning an ability to secure a greater share of benefits.

In bargaining, majority rule allows a powerful actor to form coalitions, exclude rivals, and keep the surplus (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Consensus does the opposite. By giving every participant a veto, it forces powerful states to surrender the very tools they might value most (Keohane 1984). Yet, some major international organizations, such as the GATT/WTO, are built around consensus (Steinberg 2002; Koremenos, Lipson, and Snidal 2001).

Existing answers do not fully resolve the puzzle. A common response claims that powerful states still influence outcomes through informal agenda power, even under consensus. Yet this answer is conceptually unsatisfying: under unanimity, all participants are veto players, so informal agenda influence cannot enable exclusion and surplus capture as majority rule does. More broadly, existing accounts either see consensus as constraining powerful states or as a concealment for conventional power, but they do not explain when and why a hegemon would prefer unanimity to a rule that allows exclusion.

This paper theorizes when and why a hegemon may prefer unanimity to majority rule in international organizations. The central claim is that consensus is not merely a constraint or procedural facade; rather, it is an institutional technology of power. Using Bayesian persuasion (Kamenica 2019), I show that consensus can make informational power more politically productive than the coalition arithmetic of majority rule. While the multi-receiver persuasion literature has examined information design under unanimity (Bardhi and Guo 2018), and Kim, Kim, and Van Weelden (2025) analyzes Bayesian persuasion in bilateral veto bargaining, neither offers the institutional comparison between voting rules that is central here. Thus, the key institutional choice is not between power and constraint, but between two distinct modes of exercising power.

Two forms of power are central to the analysis. *Agenda power* is the ability to control proposals, shape coalitions, and determine which alternatives reach the floor—the familiar Baron-Ferejohn logic in which the capacity to exclude is the source of surplus capture. *Informational power* is the ability to shape beliefs about the value of cooperation; in international politics, large powers typically possess deeper technical expertise and greater ability to generate authoritative interpretations of proposed rules, whereas weaker states face limited analytical capacity. This paper's focus on information *design* under different voting rules complements the literature on information *aggregation*, where Feddersen and Pesendorfer (1998) show that unanimity is the worst rule for aggregating dispersed private information. The present model reverses the direction: information is concentrated in a single actor (the hegemon), and unanimity amplifies rather than impedes its strategic value (on veto structures and bargaining power more generally, see Diermeier and Myerson 1999). Under majority rule, agenda power dominates: weak states can exclude the hegemon, so beliefs affect expected values but create no strategic discontinuity. Under unanimity, informational power dominates: every actor becomes pivotal, and the hegemon's private information creates a screening problem that generates a non-convexity in its value function.

To formalize the argument, I develop a model in which a hegemon chooses between two institutional packages that differ only in the voting rule: majority and unanimity. Proposal rights are symmetric under both rules, isolating the effect of the voting rule itself. Before weaker states decide whether to enter, the hegemon privately observes the value of cooperation and can shape their beliefs about it. The central difference is that the two rules reward this informational advantage differently. Under majority rule, weak states can form coalitions without making the hegemon pivotal, eliminating the screening problem. Under unanimity, weak states must include the hegemon in every proposal and decide how much to offer without knowing the probability of acceptance. This screening generates a cutoff at which weak proposers switch from aggressive to conservative treatment of the hegemon, creating a discontinuity in the hegemon's value function that Bayesian persuasion exploits. This creates a persuasion opportunity absent under majority, although majority may still be preferred when it facilitates participation under less optimistic priors.

The analysis yields three main results. First, I characterize the screening cutoff under unanimity and show it has no analog under majority rule, where the hegemon's value function is linear in beliefs (Propositions 1–4). Second, I show that this screening creates an additional non-convexity in the hegemon's value function, providing a persuasion opportunity absent under majority (Proposition 5). Third, I establish that the institutional comparison has a single-crossing property: there is at most one prior at which the ranking between unanimity and majority changes, and I provide a closed-form expression for the threshold (Lemma 1 and Theorem 1).

The substantive implication is that consensus is most valuable to a hegemon when the prospects of multilateral cooperation are promising enough that weaker states are willing to come to the table. Under those conditions, unanimity amplifies the hegemon's informational advantage: weaker states must include it in every deal without knowing what cooperation is truly worth, and the hegemon can shape their beliefs to exploit that uncertainty. When cooperation looks too uncertain for weaker states to participate at all, majority rule can be preferred—not because it gives the hegemon better bargaining terms, but because it lowers the bar for participation. These two forces produce a clean trade-off with a unique tipping point: above it, the hegemon prefers consensus; below it, it may prefer majority. The ranking never oscillates—the institutional comparison is not incomplete but conditional in a sharp and tractable way. Under identifiable informational conditions, consensus becomes an institutional form through which hegemonic power is exercised, not a constraint upon it.

In sum, this paper contributes to the study of international organizations in three ways. First, it offers a new mechanism explaining when and why a powerful state may rationally choose unanimity rather than just accept it. Second, it reframes institutional design as a trade-off between informational leverage under unanimity and easier participation under majority, rather than solely between power and constraint. Third, it provides a formal link between voting rules and the strategic value of private information.

The rest of the paper proceeds as follows. Section 2 presents a motivating example that previews the mechanism in a simplified three-player environment. Section 3 defines the institutional comparison; Section 4 presents the general model; Sections 5 and 6 derive the bargaining logic under majority and unanimity; Section 7 connects bargaining to entry and persuasion; Section 8 establishes the main result; Section 9 discusses extensions and scope; Section 10 concludes.

2 A Motivating Example

Before turning to the general model, consider a stripped-down environment with three players to build intuition for the mechanism. There is one hegemon H and two weak states W_1, W_2. The value of cooperation is either $V(0) = 1$ (low) or $V(1) = 2$ (high); H observes the state, the weak states do not. Each weak state's outside option is 0; H's outside option is $\alpha V(\theta)$ with $\alpha = 0.2$. A weak state is selected to make a take-it-or-leave-it offer specifying H's share y_H; if H rejects, each player receives their disagreement payoff.

Majority: exclusion eliminates screening. Under majority rule, W_1 and W_2 can pass a proposal without H's vote. The proposing weak state excludes H from the coalition; H captures $0.2V(\theta)$ from its bilateral alternative regardless of what W believes. The hegemon's expected payoff is $0.2 + 0.2\mu$, where $\mu = \Pr(\theta = 1)$—linear in beliefs, with no strategic discontinuity.

Unanimity: inclusion creates screening. Under unanimity, H's vote is required. The proposing weak state must decide how much to offer H without knowing θ:

- *Aggressive offer* ($y_H = 0.2$): type $\theta = 0$ is indifferent and accepts; type $\theta = 1$ rejects (wants 0.4). W gets 0.8 if $\theta = 0$, gets 0 if $\theta = 1$.
- *Conservative offer* ($y_H = 0.4$): both types accept. W gets 0.6 if $\theta = 0$, gets 1.6 if $\theta = 1$.

W prefers the aggressive offer when $0.8(1-\mu) > 0.6(1-\mu)+1.6\mu$, which simplifies to $\mu < 1/9$. The screening cutoff $\mu^* = 1/9$ is the belief at which W switches from aggressive to conservative treatment of H.

The jump and overpayment. Below the cutoff, H's expected payoff is the same as under majority: $0.2 + 0.2\mu$. Above the cutoff, H's expected payoff jumps to 0.4—constant, independent of μ. The source is overpayment: on the conservative branch, type $\theta = 0$ receives 0.4 but would accept 0.2. Weak states cannot avoid this cost because they must buy H's vote without knowing the type. The jump at $\mu^* = 1/9$ is 0.18, or about 16% of expected surplus at the cutoff.

Bayesian persuasion exploits the jump. The jump creates a non-convexity in H's value function that Bayesian persuasion can exploit. Suppose the prior is $p < 1/9$. Without persuasion, H's payoff under both rules is $0.2 + 0.2p$. But under unanimity, H can design a public signal that pushes W's posterior to $1/9$ with probability $9p$ (inducing conservative behavior and payoff 0.4) and to 0 with probability $1 - 9p$ (payoff 0.2). The resulting expected payoff is $0.2 + 1.8p$, strictly exceeding the majority payoff of $0.2 + 0.2p$ for every $p > 0$. Under majority, the payoff is already linear, so concavification adds nothing. This is the central asymmetry: unanimity creates a non-convexity that makes information strategically productive; majority does not.

Preview. The general model (Section 4 onwards) enriches this logic with N players, random proposer, two-round Baron-Ferejohn bargaining with discounting ($\beta < 1$), and an explicit entry stage with cost c. The richer environment produces two screening cutoffs (one per round), a richer non-convexity structure, and an entry threshold that creates a second channel through which the voting rule matters. Theorem 1 shows that the pattern here—unanimity dominates at sufficiently optimistic priors, majority can dominate only through easier entry at pessimistic priors—generalizes with a single-crossing property.

3 Institutional Comparison

The relevant comparison is between two institutional packages that differ only in the voting rule. Both packages have symmetric proposal rights: in each bargaining round, a proposer is selected uniformly at random, with each player recognized with probability $1/N$. This ensures that the comparison isolates the effect of the voting rule, holding proposal rights constant.

Definition 1 (Institutional Packages)**.** *The analysis compares two institutional packages:*

- ***Package** M **(Majority rule)**: decisions pass with strict majority support.*
- ***Package** U **(Unanimity/Consensus)**: decisions require unanimous support.*[1]

Both packages have symmetric proposal rights (random proposer with probability $1/N$). The packages differ only *in the voting rule.*

Two features of this comparison require justification.

Why symmetric proposal rights? Under either voting rule, if the hegemon were the exclusive proposer and weak states had no outside option from bargaining ($d_W = 0$), weak states' expected

[1] The model analyzes formal unanimity (every member has a veto). In practice, "consensus" at the WTO refers to the norm of not calling a formal vote, which is distinct from a formal unanimity requirement. The model's results apply most directly to settings where each member has an effective veto over outcomes, whether de jure or de facto.

bargaining payoff would be zero. No weak state would pay an entry cost $c > 0$, and the institution would never form. Monopoly proposal power is self-defeating when participation is voluntary. The relevant institutional choice is therefore between voting rules, holding proposal rights symmetric. Other forms of agenda influence—such as greater recognition probability—are discussed in the extension (Section 9.1).

Why not compare consensus with hegemonic agenda control? Because agenda control combined with any participation constraint eliminates the institution itself. The puzzle is not why a hegemon does not impose dictatorship, but why it would choose a rule that makes its own approval necessary when majority would allow others to bypass it. The relevant comparison is between majority and unanimity under the same (symmetric) proposal structure.

Weak states' disagreement payoff is normalized to $d_W = 0$. This is without loss of generality: it requires only that weak states' outside options are symmetric and strictly lower than the hegemon's. The pie $V(\theta)$ is then interpreted as the surplus above weak states' common outside option.

The hegemon's disagreement payoff is $d_H = \alpha V(\theta)$, where $\alpha \in (0, 1/r)$. This captures bilateral alternatives that are proportional to the value of multilateral cooperation—the same factors that make multilateral cooperation valuable also improve the hegemon's outside option. The parameter $\alpha < 1/r$ ensures that bilateral alternatives are strictly dominated by multilateral cooperation for all states of the world.

4 The Model

4.1 Environment

Definition 2 (Institutional Design Game)**.** *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $N-1$ weak states $W_1, \ldots, W_{N-1}$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ; the weak states do not.*

- *Disagreement payoffs: each W receives 0; H receives $\alpha V(\theta)$, with $\alpha \in (0, 1/r)$.*
- *The common discount factor is $\beta \in (0, 1)$.*
- *Each weak state that joins the institution pays an entry cost $c > 0$.*

Let $V_e(\mu) \equiv 1+\mu(r-1)$ denote the expected value of cooperation at posterior belief μ, and define $x \equiv (N-1)\alpha r$ as a notational shorthand.

4.2 Timing

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Information and entry).** The hegemon commits to a public signal structure $\pi : \{0, 1\} \to \Delta(S)$. Nature draws θ; the hegemon observes θ and the signal realization $s \in S$ is generated according to π. Weak states observe s, update to posterior $\mu = \Pr(\theta = 1 \mid s)$, and simultaneously decide whether to enter (paying cost c).
3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn bargaining game over a pie of size $V(\theta)$:
   - **Round 1**: a proposer is selected uniformly at random (probability $1/N$). The proposer offers a division. All members vote according to rule R. If accepted, payoffs are realized. If rejected, the game proceeds to Round 2 with payoffs discounted by β.
   - **Round 2 (terminal)**: a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, payoffs are realized. If rejected, each player receives their disagreement payoff.

4.3 Solution concept

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs satisfy sequential rationality.

Tie-breaking convention. When H is indifferent between accepting and rejecting an offer (offer equals reservation value), H accepts.

4.4 Information structure

Throughout the game, the hegemon's information sets are singletons: H knows θ at every decision node. Each weak state's information sets pool the two branches of θ: at any decision node, W_i knows the public signal realization and the history of proposals and responses, but not θ. After the public signal, weak states hold posterior μ over θ. This asymmetry is what generates the screening problem when a weak state proposes under unanimity.

Figures 1 and 2 present the extensive form of the game: Figure~1 covers institutional choice, Bayesian persuasion, and entry (Stages 0–1); Figure~2 details the bargaining subgame under unanimity (Stage 2).

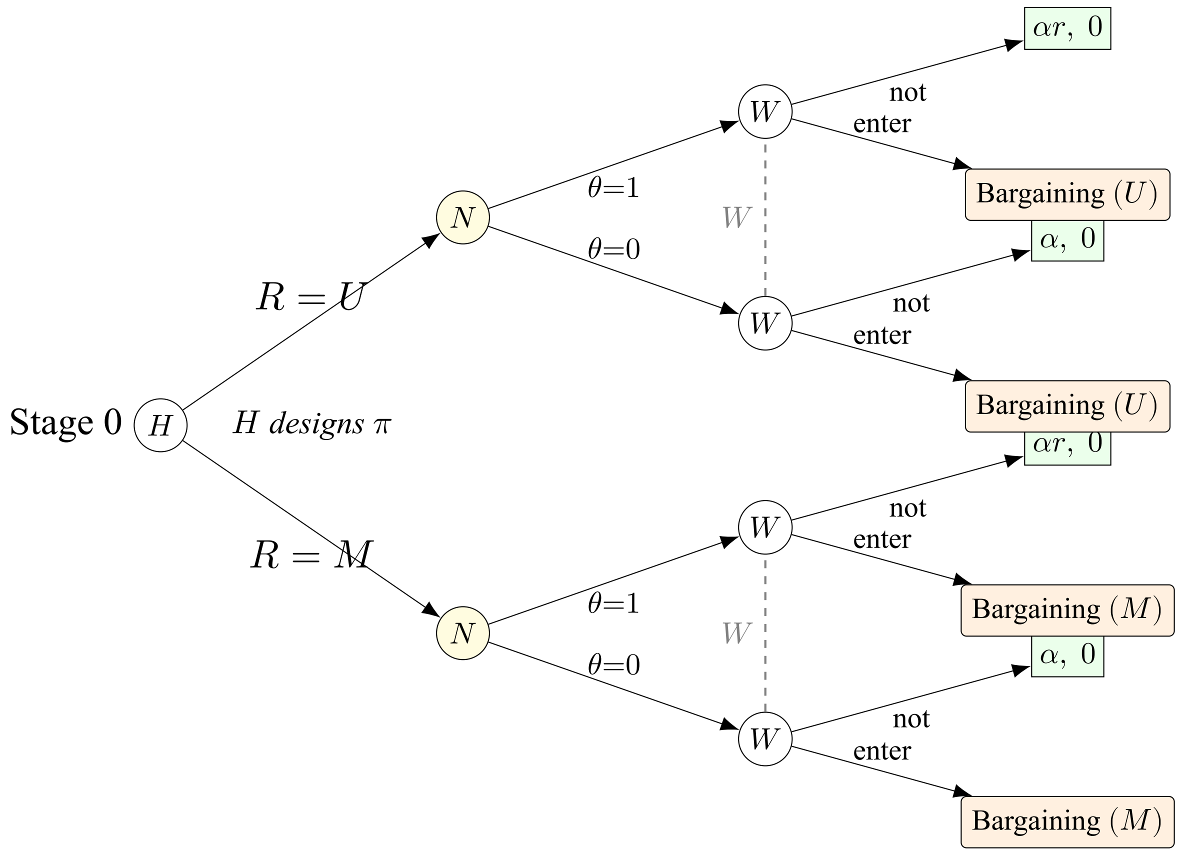

Figure 1: Extensive form: Stages 0–1 (institutional choice, information design, and entry). At Stage 0, H selects voting rule $R \in \{M, U\}$ and commits to a Bayesian persuasion signal π. Nature draws $\theta \in \{0, 1\}$; weak states observe signal realization s, form posterior $\mu = \Pr(\theta{=}1 \mid s)$, and decide entry (cost c). Dashed lines: W's information sets (W does not observe θ). Non-entry payoffs: $(u_H, u_W) = (\alpha V(\theta),\ 0)$.

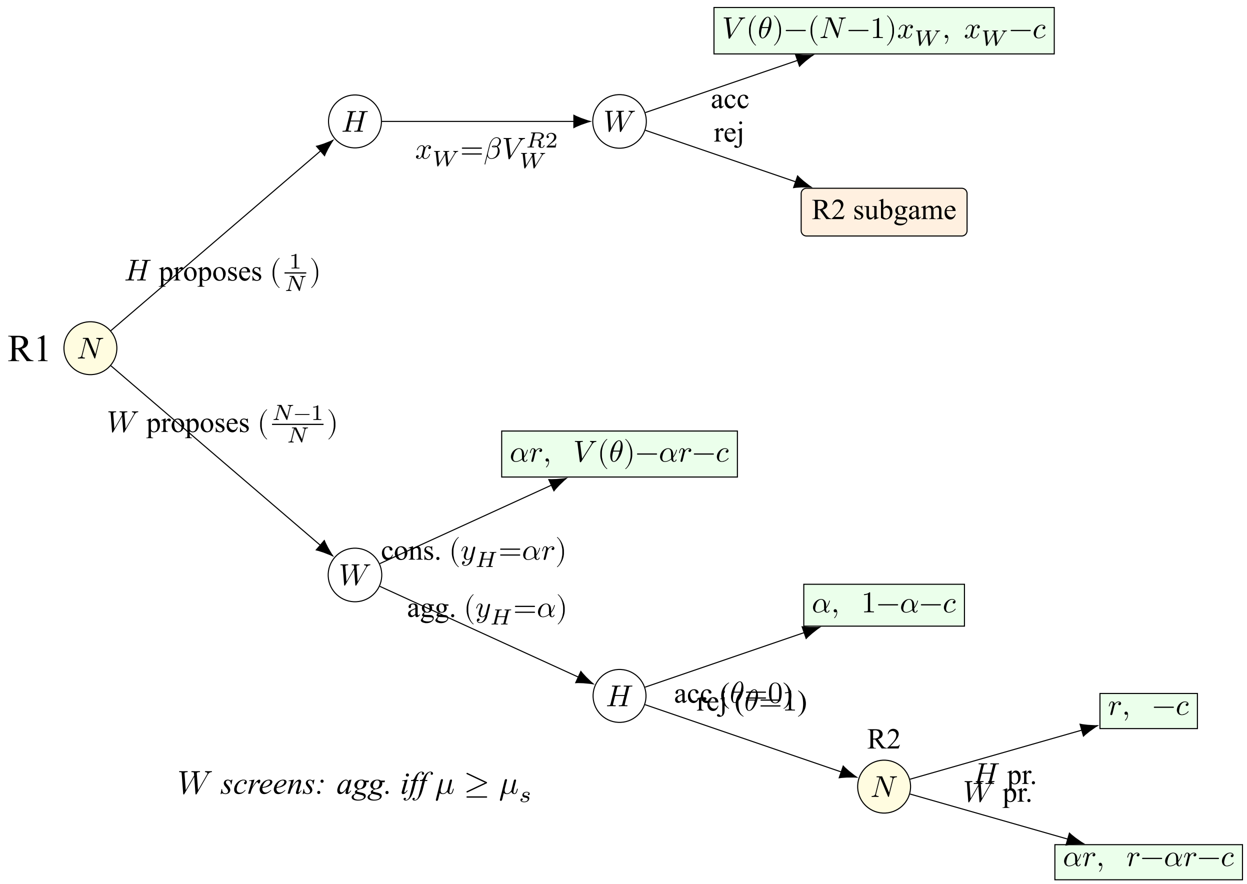

Figure 2: Extensive form: Stage 2 (bargaining under unanimity, Rounds 1–2). Payoffs are (u_H, u_W). A proposer is selected uniformly ($1/N$). When W proposes, W chooses between aggressive ($y_H = \alpha$) and conservative ($y_H = \alpha r$). Under the aggressive offer, type $\theta{=}0$ accepts (tie-breaking) and $\theta{=}1$ rejects, revealing the state and leading to R2 (payoffs discounted by β). In R2, W knows $\theta{=}1$ (complete information); W offers αr, H offers 0; disagreement yields $(\alpha r, -c)$. When H proposes in R1, H offers βV_W^{R2} to each W; W accepts in equilibrium. Under majority rule, H's vote is not pivotal, eliminating the screening mechanism (Proposition 1).

5 Majority Rule: Linear Bargaining Without Screening

Under majority rule, a proposal passes if it receives at least $q = \lfloor N/2 \rfloor + 1$ votes. The key structural feature is that weak states can form a winning coalition without including the hegemon. When W proposes, W need not secure H's vote—W can assemble a majority from the other weak states alone. This eliminates the screening problem.

5.1 Terminal round (R2) under majority

When H proposes (prob $1/N$): H needs $q-1$ votes from weak states. Since $d_W = 0$, each W accepts any non-negative offer. H offers 0 to each of $q-1$ weak states and keeps $V(\theta)$.

When W_j proposes (prob $1/N$): W_j excludes H from the winning coalition and assembles a majority from the other $q-1$ weak states (each offered 0). The hegemon, excluded from the proposal, captures $\alpha V(\theta)$ from its bilateral alternative. The proposing weak state keeps $V(\theta) - \alpha V(\theta) = (1-\alpha)V(\theta)$.

Since H is excluded and captures $\alpha V(\theta)$ regardless of W's beliefs, there is no screening problem: H's payoff does not depend on the posterior μ.

The R2 continuation values are:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1+(N-1)\alpha]}{N}, \tag{1}$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N}. \tag{2}$$

Both are affine in μ. No screening, no jump.

5.2 First round (R1) under majority

When H proposes in R1: H offers $\beta V_W^{R2}(\mu, M)$ to each of $q-1$ weak states (their discounted R2 continuation). When W proposes in R1: W excludes H from the winning coalition and buys $q-1$ weak states at their discounted R2 continuation value. The hegemon, excluded, receives $\alpha V(\theta)$ from its bilateral alternative.

The key point: everything remains linear in μ.

Proposition 1 (Majority rule produces no screening)**.** *Under majority rule with symmetric proposal rights, the hegemon's expected continuation value $E_\theta[V_H^{R1}(\theta, \mu, M)]$ is affine in posterior beliefs μ. There is no screening cutoff. Weak states' behavior does not change discretely as beliefs cross any threshold.*

Proof. Under majority, W's proposal passes regardless of H's vote. Therefore W never faces a choice between offers conditional on H's acceptance. The transfer to H is type-specific ($\alpha V(\theta)$) regardless of beliefs μ. Since all continuation values are affine functions of $V(\theta)$, and $E[V(\theta)] = V_e(\mu)$ is affine in μ, the hegemon's expected payoff is affine in μ. □

The intuition is straightforward. Majority transforms bargaining into coalition arithmetic. The hegemon's private information affects the value of the agreement, but it does not create a strategic discontinuity. There is no point at which weak states must change their behavior because they cannot identify H's type. Under majority, H's type is irrelevant to whether proposals pass. Bayesian persuasion under majority therefore operates only through the entry channel.

6 Consensus: Screening Through Pivotal Inclusion

Under unanimity, every proposal requires the approval of all members, including the hegemon. This changes the bargaining logic fundamentally. When a weak state proposes, it must decide how much to offer H without knowing whether the pie is high or low. That decision is a screening problem.

6.1 Terminal round (R2) under unanimity

When H proposes (prob $1/N$): H must satisfy all weak states. Since $d_W = 0$, H offers 0 to each W. All W's accept (indifferent). H keeps $V(\theta)$.

When W_j proposes (prob $1/N$): W_j offers 0 to other W's (accepted by indifference). W_j must decide what to offer H. By dominance, only two offers are relevant:

- **Aggressive offer** ($y_H = \alpha$): satisfies type $\theta = 0$ (indifferent: $\alpha = \alpha V(0)$), but rejected by type $\theta = 1$ (since $\alpha < \alpha r = \alpha V(1)$).

- **Conservative offer** ($y_H = \alpha r$): satisfies both types (type $\theta = 0$ strictly prefers $\alpha r > \alpha$; type $\theta = 1$ is indifferent).

The screening cutoff in R2 solves W's indifference between the two offers:

$$\mu_s^{R2} = \frac{\alpha(r-1)}{r-\alpha}. \tag{3}$$

For $\mu < \mu_s^{R2}$, W plays aggressive. For $\mu > \mu_s^{R2}$, W plays conservative.

The R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1 + (N-1)\alpha]}{N} \quad \text{(constant in } \mu\text{)}, \tag{4}$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1 + (N-1)\alpha}{N}, \tag{5}$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1 + (N-1)\alpha r}{N} \quad \text{(overpaid)}. \tag{6}$$

Proposition 2 (Overpayment under unanimity)**.** *Under unanimity on the conservative branch ($\mu > \mu_s^{R2}$), type $\theta = 0$ receives strictly more than under majority:*

$$\frac{1 + (N-1)\alpha r}{N} > \frac{1 + (N-1)\alpha}{N}.$$

This overpayment arises because W cannot tailor the offer to the realized type. On the aggressive branch, payoffs are identical across rules. Type $\theta = 1$ receives the same under both rules for all μ.

This is the source of the mechanism. Under majority, W bypasses H, so H always receives the type-specific payoff $\alpha V(\theta)$. Under unanimity, W must include H but cannot distinguish types. On the conservative branch, W offers αr to *both* types, overpaying type $\theta = 0$ by $(N-1)\alpha(r-1)/N$ relative to majority.

6.2 First round (R1) under unanimity

When H proposes in R1 (prob $1/N$): H offers $\beta V_W^{R2}(\mu)$ to each W. Both types make the same offer (pooling). All W's accept. No information is revealed.

When W_j proposes in R1 (prob $1/N$): W_j faces the screening problem again, now with R2 as the continuation game. The two relevant offers are:

- **Conservative offer**: $y_H^{con} = \beta(r + x)/N$, the discounted R2 value of the high type. Both types accept; the game ends in R1.
- **Aggressive offer**: $y_H^{agg} = \beta(1 + x)/N$, the discounted R2 value of the low type (under the belief that will prevail after rejection). Type $\theta = 0$ accepts (indifferent). Type $\theta = 1$ rejects; the game moves to R2 with W knowing $\theta = 1$.

The R1 screening cutoff solves W's indifference between these strategies. From the technical derivation (Appendix A), the cutoff depends on a piecewise quadratic equation.

Proposition 3 (R1 screening cutoff under consensus)**.** *Under unanimity, there exists a unique $\mu_s^{R1} \in (0, 1)$ such that the weak proposer prefers the aggressive offer for $\mu < \mu_s^{R1}$ and the conservative offer for $\mu > \mu_s^{R1}$.*

When $\mu_s^{R1} > \mu_s^{R2}$ (which holds when $\Delta_1(\mu_s^{R2}) > 0$, i.e., when α is small enough that the R1 cutoff lies on the conservative R2 branch), the cutoff is given by:

$$\mu_s^{R1} = \frac{\phi - \sqrt{\phi^2 - 4(N-2)}}{2(N-2)}, \qquad \phi \equiv \frac{rN - \beta(N - 1 + r)}{\beta(r-1)}. \tag{7}$$

In this regime, the cutoff does not depend on α.

Proof. Existence follows from boundary conditions: the gain from aggressive is positive at $\mu = 0$ ($\Delta_1(0) = \beta(r-1)/N > 0$) and negative at $\mu = 1$ ($\Delta_1(1) = r(\beta-1) < 0$). Uniqueness follows from convexity of the relevant quadratic on the high branch (see Appendix A). The α-independence in the principal case arises from exact cancellation: on the conservative branch of V_W^{R2}, the terms involving α in the aggressive-vs-conservative comparison cancel algebraically. $\square$

The α-independence of the R1 cutoff in the main regime (Case 2) is substantively important. The screening mechanism does not require that H have a particularly strong outside option. What matters is the informational rent $(r - 1)$, the patience β, and the coalition structure N.

For $\beta = 1$, the cutoff simplifies to $\mu_s^{R1} = 1/(N - 2)$.

6.3 The jump in the hegemon's expected payoff

Proposition 4 (Screening creates an informational rent)**.** *Under unanimity, the hegemon's expected continuation payoff in R1 has an upward jump at* μ_s^{R1}*. The magnitude of the jump is:*

$$\textit{Jump} = (1 - \mu_s^{R1}) \cdot \frac{(N-1)\beta(r-1)}{N^2} > 0. \tag{8}$$

Proof. Below μ_s^{R1}, type $\theta = 0$ receives $\beta(1+x)/N$ when W proposes (aggressive: accepted at reservation). Above μ_s^{R1}, type $\theta = 0$ receives $\beta(r+x)/N$ (conservative: overpaid). The difference is $\beta(r-1)/N$, weighted by prob $(N-1)/N$ of W proposing and by $(1-\mu_s^{R1})$ for the probability that $\theta = 0$. □

The intuition: at the screening cutoff, weak proposers switch from treating the hegemon as certainly the low type (aggressive) to hedging against the possibility of the high type (conservative). The conservative offer pays the low type as if it could be the high type. This overpayment creates a discrete upward shift in the hegemon's expected payoff. The jump is larger when:

- r is large (greater asymmetry between types),
- β is large (more patient players make the continuation threat more credible),
- μ_s^{R1} is small (larger probability mass on the conservative region),
- N is moderate (the per-player jump is $(N-1)/N^2$, maximized at intermediate N).

Example 1 (Screening jump: concrete magnitudes)**.** *Let* $N = 5$*,* $r = 1.5$*,* $\alpha = 0.3$*,* $\beta = 0.9$*. The R1 screening cutoff is* $\mu_s^{R1} \approx 0.197$*. At beliefs just below the cutoff, weak proposers play aggressively and the hegemon's expected continuation payoff is* $E[V_H^{R1}] \approx 0.544$ *(computed from equations* (4)–(6) *and the R1 expressions in Section*

refR1consensus). At beliefs just above it, weak proposers switch to conservative offers and the payoff jumps to $E[V_H^{R1}] \approx 0.602$*—a discrete increase of* 0.058*, or about* 5.3% *of expected surplus* $V_e(\mu)$*. Under majority rule at the same belief,* $E[V_H^{R1}] \approx 0.428$*, with no jump at any belief. The screening mechanism thus gives the hegemon* 27% *more than majority on the aggressive branch and* 41% *more on the conservative branch, with the discrete gap between the two branches providing the non-convexity that Bayesian persuasion exploits.*

7 Entry and Bayesian Persuasion

7.1 Entry decisions

A weak state enters if its expected bargaining payoff exceeds the entry cost:

$$V_W^{R1}(\mu, R) \geq c.$$

The entry cost c admits several interpretations in the IO context. It may represent the fixed costs of negotiating accession (as with WTO accession), the per-round costs of preparing delegations and engaging substantively in negotiations, or the political costs of committing to multilateral disciplines. The model is most applicable to settings where participation is a genuine decision at each negotiation—as in trade rounds, where states choose how actively to engage—rather than to settings where membership is automatic and exit is prohibitive. In established organizations, c is best interpreted as the cost of *substantive engagement*: allocating scarce diplomatic and analytical resources to a particular negotiating round.

The entry threshold $\tau(R)$ is the posterior at which W is indifferent.

Under majority, $V_W^{R1}(\mu, M)$ is affine in μ, so the entry threshold is unique and has a simple closed-form expression.

Under unanimity, $V_W^{R1}(\mu, U)$ is piecewise affine (reflecting the screening regimes), increasing in μ on each branch. The entry threshold $\tau(U)$ solves $V_W^{R1}(\tau(U), U) = c$. Because V_W^{R1} has different functional forms on each side of the R1 and R2 cutoffs, a single closed-form expression for $\tau(U)$ is unavailable. On each branch, however, the threshold has a closed form: for instance, if $\tau(U)$ falls on the conservative R1 branch, $\tau(U) = [N^2c - N + \beta(r - 1 + Nr\alpha)]/[(N + \beta)(r - 1)]$. The branch on which $\tau(U)$ lies is determined by checking whether the solution falls within the relevant interval.

7.2 Value functions

The hegemon's reduced-form value function maps posteriors to expected payoffs:

$$v(\mu, R) = \begin{cases} 0 & \text{if } V_W^{R1}(\mu, R) < c \\ E_\theta[V_H^{R1}(\theta, \mu, R)] & \text{if } V_W^{R1}(\mu, R) \geq c \end{cases}$$

Under majority: $v(\mu, M)$ is zero below $\tau(M)$ and affine above it. The value function has a single non-convexity: the entry threshold. Bayesian persuasion under majority can only exploit the entry channel.

Under unanimity: $v(\mu, U)$ has two sources of non-convexity: 1. The entry threshold at $\tau(U)$: value jumps from zero to positive. 2. The screening cutoff at μ_s^{R1}: value jumps upward due to the switch from aggressive to conservative treatment.

Proposition 5 (Unanimity creates an additional persuasion opportunity)**.** *The value function $v(\mu, U)$ has a screening non-convexity at μ_s^{R1} that is absent in $v(\mu, M)$. For posteriors near the screening cutoff, Bayesian persuasion under unanimity generates strictly higher payoffs than under majority (holding entry conditions constant).*

Proof. Under majority, $v(\mu, M)$ is affine for $\mu \geq \tau(M)$ (Proposition 1), hence concave on that domain. Concavification adds nothing beyond the entry threshold. Under unanimity, the upward jump at μ_s^{R1} (Proposition 4) means $v(\mu, U)$ is not concave on $[\tau(U), 1]$. The concave envelope therefore lies strictly above $v(\mu, U)$ for a range of posteriors near μ_s^{R1}, creating persuasion gains absent under majority. □

7.3 Optimal persuasion

By the standard Bayesian persuasion result (Kamenica and Gentzkow 2011), the hegemon's optimal payoff under rule R equals the concavification of $v(\mu, R)$ evaluated at the prior p:

$$\Pi_H^*(R) = \text{cav}\, v(p, R).$$

Under majority, the concavification is standard: if $p < \tau(M)$, the optimal signal induces posteriors at 0 and $\tau(M)$, effectively choosing whether to induce entry or not.

Under unanimity, the concavification can exploit both entry and screening. The optimal signal can induce posteriors that straddle the screening cutoff, placing mass on both sides of μ_s^{R1}. This dual exploitation is the paper's distinctive contribution. Figure **??** illustrates the value functions and their concavifications under both rules.

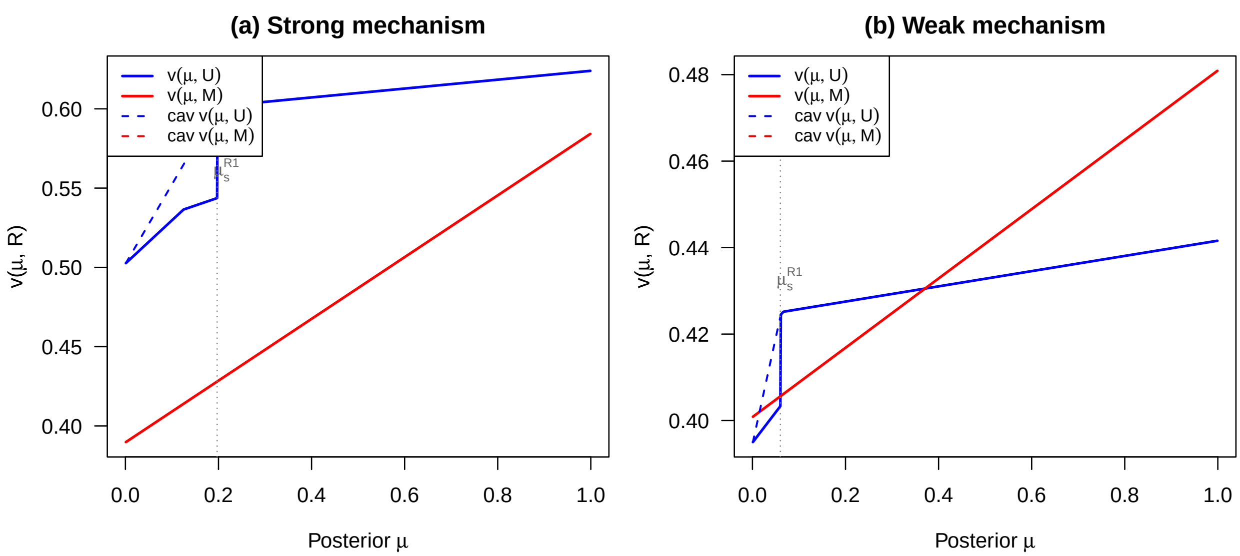

Figure 3: (#fig:bp_illustration)Value functions and concavification under majority (red) and unanimity (blue). Dashed lines show concave envelopes. Left panel (a): baseline parameters ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$, $c = 0.1$) where the screening jump under unanimity creates a large concavification gain. Right panel (b): parameters near the boundary of the unanimity-preferred region ($N = 5$, $r = 1.2$, $\alpha = 0.3$, $\beta = 0.7$, $c = 0.1$) where the mechanism is weaker and the concavification gains are smaller.

8 Institutional Choice

This section establishes the paper's main result: conditions under which the hegemon prefers unanimity.

8.1 Conditional payoff comparison

Before turning to the full institutional comparison with persuasion, I establish a result that is central to the analysis: conditional on the institution forming, unanimity gives the hegemon a strictly higher expected continuation payoff than majority for all interior beliefs.

Lemma 1 (Conditional payoff dominance of unanimity)**.** *Define the threshold*

$$\alpha^*(N,\beta) \equiv \frac{\beta(q-1)}{N(N-1)-\beta(N^2-N-q+1)}, \qquad q = \lfloor N/2 \rfloor + 1.$$

For $\alpha < \alpha^*(N,\beta)$ and every $\mu \in (0,1]$,

$$E_\theta[V_H^{R1}(\theta,\mu,U)] > E_\theta[V_H^{R1}(\theta,\mu,M)].$$

Proof. Let $P \equiv \beta(q-1)(1-\alpha)$, $Q \equiv N(N-1)\alpha$, and $x = (N-1)\alpha r$. Under majority, $E[V_H^{R1}(\mu,M)] = \lambda_M V_e(\mu)$. Define the backbone:

$$D_{\text{base}}(\mu) \equiv \frac{(P-Q)V_e(\mu) + Q\beta r}{N^2},$$

and two correction terms:

$$\delta_{R2}(\mu) \equiv \frac{(N-1)\beta[\mu(r-\alpha)-\alpha(r-1)]}{N^2}, \qquad \delta_{R1}(\mu) \equiv \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}.$$

Direct computation (Appendix B.5) shows that the difference $D(\mu) \equiv E[V_H^{R1}(\mu,U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\text{R1 conservative}\} \cdot \delta_{R1}(\mu).$$

The corrections have two key properties: $\delta_{R2}(\mu_s^{R2}) = 0$ exactly (ensuring continuity at the R2 cutoff), and $\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$ (the R1 conservative correction only adds positivity).

Step 1: $D_{base} > 0$ on $[0,1]$. D_{base} is affine. At $\mu = 1$: $D_{\text{base}}(1) = r[P - Q(1-\beta)]/N^2$. Setting $P - Q(1-\beta) = 0$ yields $\alpha = \alpha^*$; since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$. At $\mu = 0$: $D_{\text{base}}(0) =$

$(P - Q + Q\beta r)/N^2$. One shows (Appendix B.5) that $D_{\text{base}}(0) > D_I(0) > 0$ under $\alpha < \alpha^*$, where $D_I(0)$ is the full difference at zero (which involves a threshold $\bar{\alpha}_0 > \alpha^*$). Affine with positive endpoints: $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 2: Assembly. On Region II (conservative R2, aggressive R1: $\mu \in (\mu_s^{R2}, \mu_s^{R1})$): $D = D_{\text{base}} > 0$. On Region I (aggressive R2: $\mu < \mu_s^{R2}$): $D = D_{\text{base}} + \delta_{R2}$, affine between $D_I(0) > 0$ and $D_{\text{base}}(\mu_s^{R2}) > 0$. On Region III (conservative R1: $\mu > \mu_s^{R1}$): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$. At the R1 cutoff, the jump $\delta_{R1}(\mu_s^{R1}) > 0$ ensures D increases discretely. □

Lemma 1 shows that, when the hegemon's outside option is not too strong ($\alpha < \alpha^*$), unanimity gives the hegemon a strictly higher continuation payoff than majority conditional on the institution forming. The proof works by decomposing the payoff difference into a backbone term D_{base} that is affine and universally positive under the condition on α, plus two corrections that either vanish at their activation point (δ_{R2}) or are non-negative everywhere (δ_{R1}). The backbone captures two forces: when the hegemon proposes, unanimity reduces the weak states' continuation values, allowing cheaper agreement; when a weak state proposes, unanimity makes the hegemon pivotal, forcing compensation above the bare outside option. The threshold α^* decreases with N: as the organization grows, unanimity requires buying more votes, and the cost can offset the pivotal-inclusion advantage unless α is sufficiently small. For the empirically relevant parameter range, majority therefore never dominates through conditional bargaining payoffs. Any advantage of majority must arise exclusively through the entry margin.

8.2 The comparison without persuasion

Without Bayesian persuasion (i.e., with no signal), weak states enter only if $V_W^{R1}(p, R) \geq c$. Under majority, the budget identity gives $V_W^{R1}(\mu, M) = [V_e(\mu) - E[V_H^{R1}(\mu, M)]]/(N - 1)$. Under unanimity, the budget identity holds exactly on the conservative R1 branch and with inequality $E[V_H] + (N - 1)V_W \leq V_e(\mu)$ on the aggressive branch (due to surplus destruction from discounting when $\theta = 1$ rejects). This means $V_W^{R1}(\mu, U) \leq [V_e(\mu) - E[V_H^{R1}(\mu, U)]]/(N - 1)$. Since Lemma 1 establishes $E[V_H^{R1}(\mu, U)] > E[V_H^{R1}(\mu, M)]$, it follows a fortiori that $V_W^{R1}(\mu, M) > V_W^{R1}(\mu, U)$ for $\mu < 1$. To see this: $V_W^{R1}(\mu, U) \leq [V_e(\mu) - E[V_H^{R1}(\mu, U)]]/(N - 1) < [V_e(\mu) - E[V_H^{R1}(\mu, M)]]/(N - 1) = V_W^{R1}(\mu, M)$, where the first inequality uses the budget identity under unanimity and the second uses Lemma 1. So entry is always easier under majority: $\tau(U) \geq \tau(M)$.

Without persuasion:

- If entry occurs under both rules ($p \geq \tau(U)$), unanimity dominates by Lemma 1.
- If entry occurs only under majority ($\tau(M) \leq p < \tau(U)$), majority trivially dominates.

8.3 The comparison with persuasion

With Bayesian persuasion, the hegemon's payoff under each rule is cav $v(p, R)$. The institutional choice reduces to comparing these concavified values.

To state the theorem, define the majority payoff coefficient

$$\lambda_M \equiv \frac{N[1+(N-1)\alpha] - \beta(q-1)(1-\alpha)}{N^2},$$

so that $v(\mu, M) = \lambda_M V_e(\mu)$ whenever entry occurs. Let

$$\kappa_M \equiv \frac{(1-\alpha)\big[N(N-1)+\beta(q-1)\big]}{N^2(N-1)}, \qquad \tau(M) = \max\left\{0,\ \frac{c/\kappa_M - 1}{r-1}\right\}.$$

Under unanimity, let $\tau(U)$ denote the entry threshold and define

$$S_U \equiv \max_{\mu \geq \tau(U)} \frac{v(\mu, U)}{\mu}.$$

If $\tau(M) > 0$, let $S_M \equiv v(\tau(M), M)/\tau(M)$.

Theorem 1 (Threshold prior for institutional choice)**.** *Suppose $\alpha < \alpha^*(N, \beta)$. The comparison between* cav $v(p, U)$ *and* cav $v(p, M)$ *has the following structure.*

(a) *If $\tau(U) = 0$, then unanimity strictly dominates for every $p \in (0, 1)$.*

(b) *If $\tau(M) = 0 < \tau(U)$, there exists a unique threshold*

$$p^* = \frac{\lambda_M}{S_U - \lambda_M(r-1)} \in (0, \tau(U))$$

such that majority dominates for $p < p^*$ and unanimity dominates for $p > p^*$.

(c) *If $0 < \tau(M) \le \tau(U)$ and $S_U > S_M$, then unanimity strictly dominates for every $p \in (0,1)$.*

(d) *If $0 < \tau(M) \le \tau(U)$ and $S_U \le S_M$, there exists a unique threshold*

$$p^* = \frac{\lambda_M}{S_U - \lambda_M(r-1)} \in [\tau(M), \tau(U)]$$

such that majority weakly dominates for $p < p^*$ and unanimity strictly dominates for $p > p^*$.

In particular, the institutional comparison has the single-crossing property*: there is at most one prior at which the ranking changes.*

Proof. Step 1: High priors ($p \ge \tau(U)$). Entry occurs under both rules. By Lemma 1, $v(p, U) > v(p, M)$. Under majority, $v(\mu, M)$ is affine above $\tau(M)$, hence already concave, so $\operatorname{cav} v(p, M) = v(p, M)$. Therefore $\operatorname{cav} v(p, U) \ge v(p, U) > v(p, M) = \operatorname{cav} v(p, M)$. This proves part (a) when $\tau(U) = 0$.

Step 2: Concavification below $\tau(U)$. Assume $\tau(U) > 0$. Since $v(\mu, U) = 0$ for $\mu < \tau(U)$ and $v(0, U) = 0$, the concavification on $[0, \mu_U^*]$ (where μ_U^* achieves S_U) is the line $\operatorname{cav} v(p, U) = S_U \cdot p$. Under majority: if $\tau(M) = 0$, then $\operatorname{cav} v(p, M) = \lambda_M V_e(p)$; if $\tau(M) > 0$, then $\operatorname{cav} v(p, M) = S_M \cdot p$ for $p < \tau(M)$ and $\operatorname{cav} v(p, M) = \lambda_M V_e(p)$ for $p \ge \tau(M)$.

Step 3: Case $\tau(M) = 0 < \tau(U)$. For $p \in (0, \tau(U))$, $D(p) = S_U \cdot p - \lambda_M V_e(p) = [S_U - \lambda_M(r-1)]p - \lambda_M$, which is linear. $D(0) = -\lambda_M < 0$; by Step 1, $D(\tau(U)) > 0$. So D has a unique zero $p^* = \lambda_M/[S_U - \lambda_M(r-1)] \in (0, \tau(U))$. (The denominator $S_U - \lambda_M(r-1)$ is necessarily positive because $D(p)$ is linear with $D(0) < 0$ and $D(\tau(U)) > 0$, so the slope must be positive.) This proves (b).

Step 4: Case $0 < \tau(M) \le \tau(U)$, $S_U > S_M$. For $p < \tau(M)$: $\operatorname{cav} v(p, U) = S_U p > S_M p = \operatorname{cav} v(p, M)$. For $p \in [\tau(M), \tau(U))$: $D(p) = [S_U - \lambda_M(r-1)]p - \lambda_M$ is linear with $D(\tau(M)) = (S_U - S_M)\tau(M) > 0$ and $D(\tau(U)) > 0$ by Step 1. For $p \ge \tau(U)$: Step 1 applies. This proves (c).

Step 5: Case $0 < \tau(M) \le \tau(U)$, $S_U \le S_M$. For $p < \tau(M)$: $\operatorname{cav} v(p, U) = S_U p \le S_M p$. For $p \in [\tau(M), \tau(U))$: $D(\tau(M)) \le 0$ and $D(\tau(U)) > 0$, so D crosses zero once at $p^* \in$

$[\tau(M), \tau(U)]$. For $p \geq \tau(U)$: Step 1 applies. This proves (d).

Since each case admits at most one crossing, the comparison has the single-crossing property. $\square$

Remark 1 (Interpretation)**.** *The theorem provides a closed-form threshold p^* that separates priors at which majority dominates from those at which unanimity dominates. The threshold exists only when entry is harder under unanimity ($\tau(U) > 0$). When it exists, majority dominates at low priors (where entry is the binding constraint) and unanimity dominates at high priors (where the screening advantage takes effect). The single-crossing property means that there is no regime in which the ranking oscillates.*

Remark 2 (Consensus is most valuable when informational power substitutes for agenda power)**.** *In cases (a) and (c), unanimity dominates globally. Case (c) requires $S_U > S_M$, which is more likely when: (i) r is large (greater informational asymmetry), (ii) β is large (more patient bargaining), and (iii) N is moderate. The advantage of unanimity increases when the screening jump is large relative to the linear gain under majority.*

8.4 Numerical characterization

Figure **??** shows that the hegemon prefers unanimity across a broad range of parameters when the prior is low ($p = 0.05$). In these regions, cases (a) or (c) of Theorem 1 apply: unanimity dominates globally. The unanimity-dominance region is larger for higher r (greater informational asymmetry) and higher β (more patient bargaining). In the red regions, case (b) or (d) applies and the threshold prior p^* lies above 0.05, so majority dominates at this particular prior but unanimity dominates at higher priors.

9 Discussion

9.1 Partial Agenda Influence and Informal Power

The baseline model uses symmetric proposal rights ($p_H = 1/N$) to isolate the effect of the voting rule. A natural objection is that powerful states exercise informal agenda influence—through control over drafting, chairing of meetings, or greater diplomatic salience—so the symmetric base-

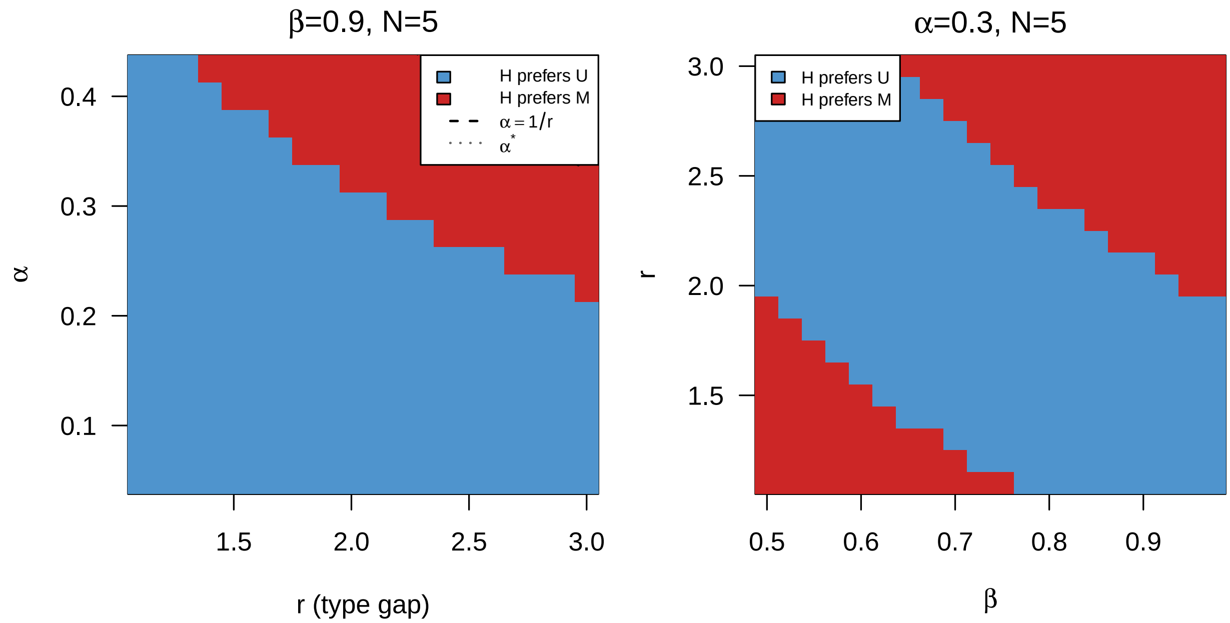

Figure 4: (#fig:parameter_regions)Parameter regions where the hegemon prefers unanimity (blue) vs majority (red) at prior $p = 0.05$, restricted to $\alpha < \alpha^*(N, \beta)$ (the domain where Lemma 1 applies). Left: r vs α plane with $\beta = 0.9$, $N = 5$, $c = 0.1$. Right: β vs r plane with $\alpha = 0.3$, $N = 5$, $c = 0.1$.

line is unrealistic. Appendix C extends the model to allow H to be recognized with probability $p_H > 1/N$, while each W is recognized with probability $p_W = (1 - p_H)/(N - 1)$.

The extension reveals that consensus and informal agenda influence are not substitutes—they interact non-monotonically. Under majority, the value function remains linear in beliefs regardless of p_H: no screening, no jump. Under unanimity, the screening cutoff μ_s^{R2} is unchanged, but the jump magnitude becomes $(1 - \mu_s^{R2})(1 - p_H)\alpha(r - 1)$, decreasing in p_H. In R1, three effects operate simultaneously: a frequency effect (W proposes less often, reducing screening opportunities), a gap effect (H's higher continuation value widens the spread between aggressive and conservative offers), and a cutoff effect (the screening threshold rises). The net R1 jump scales with $p_H(1-p_H)$ holding the cutoff fixed, producing a hump-shaped relationship.

Proposition 6 (Non-monotonic effect of agenda influence)**.** *Under unanimity, the R1 screening jump as a function of recognition probability p_H is:*

$$J(p_H) = (1 - \mu_s^{R1}(p_H))(1 - p_H)p_H\beta(r - 1).$$

Holding μ_s^{R1} fixed, J is maximized at $p_H = 1/2$. In the full model, μ_s^{R1} increases with p_H, shifting the peak below $1/2$. Symmetric proposal rights ($p_H = 1/N$) can therefore be understood as participation-preserving restraint rather than as absence of hegemonic influence.

9.2 Illustration: The GATT/WTO

The GATT/WTO illustrates the interaction between consensus and informal agenda influence. Since 1947, the multilateral trading system has operated predominantly by consensus, even though formal voting provisions exist (Steinberg 2002). The United States and other major trading powers were central to the system's creation and could plausibly have imposed majority rule.

The model's informational asymmetry assumption finds support in the trade context. Evaluating the effects of tariff schedules, anticipating distributive consequences, and assessing long-term implications of rule changes require sustained investment in analytical infrastructure. Major trading powers maintain large permanent delegations with deep sectoral expertise; many developing country delegations operate with far smaller teams, particularly during complex technical phases of negotiations. This capacity asymmetry means that weaker states often face genuine uncertainty about the value of proposed agreements.

In practice, major trading powers exercise informal agenda influence through the Green Room process, control over drafting, and technical expertise. The partial agenda influence extension provides the framework for interpreting this: moderate informal influence can amplify the screening channel—it increases the stakes when W does get to propose—while monopoly agenda control would eliminate participation, destroying the institution. The persistence of consensus is consistent with the coexistence of informal influence and formal equality.

The model's mechanism explains a pattern that existing accounts describe but do not fully explain: powerful states in the WTO appear to benefit from consensus despite lacking formal agenda control. The mechanism is not merely that informal practices survive consensus, but that consensus creates the strategic environment—pivotal inclusion under uncertainty—in which informational power becomes most valuable. The model does not claim that the GATT/WTO was designed for this reason, or that all aspects of WTO politics reduce to this mechanism. It shows that consensus and informational asymmetry interact in a way that can benefit powerful states, offering one explanation for why hegemonic preference for consensus is not paradoxical.

The informational mechanism generates a distinctive observable implication: the hegemon's advantage from consensus should be largest in negotiations where informational asymmetry is substantial and where weak states lack independent analytical capacity. In the WTO context, this suggests that consensus should matter most in complex regulatory areas—services, intellectual property, investment—where evaluating proposals requires deep technical expertise, and least in areas where distributive consequences are transparent, such as linear tariff cuts. This prediction distinguishes the informational mechanism from legitimacy accounts (which predict consensus matters uniformly) and from self-enforcement accounts (which predict consensus matters most when enforcement is weak, regardless of information).[2]

9.3 Scope

When consensus favors the hegemon. The mechanism requires that: (i) informational asymmetry is relevant (H knows θ, W does not); (ii) the hegemon has a valuable outside option ($\alpha > 0$); (iii) entry is costly and belief-dependent; (iv) the voting rule makes H pivotal.

The model assumes the hegemon can commit to an information structure before observing θ. This is a strong assumption that warrants discussion. Three considerations support its plausibility in the IO context. First, *reputation*: in repeated interactions, a hegemon that deviates from an announced disclosure policy suffers credibility costs in future negotiations, which can sustain commitment in equilibrium. Second, *institutional transparency rules*: organizations like the WTO have formal provisions for notifications, trade policy reviews, and dispute settlement that constrain and channel information disclosure, serving as partial commitment devices. Third, *upper bound interpretation*: the Bayesian persuasion payoff represents the maximum value of the hegemon's informational advantage. Even if full commitment is unavailable, the qualitative institutional comparison—

[2] The $K > 2$ extension in Appendix D provides a formal basis for this implication. As the number of relevant states increases—corresponding to negotiations that bundle multiple issue dimensions under a Single Undertaking (intellectual property, services, industrial goods, agriculture)—the parametric threshold α_3^* tightens: the hegemon needs a weaker outside option for unanimity to dominate. In the model, higher K dilutes informational power because the screening problem becomes more complex, with correction terms that can work against H (see Section D.5). This is consistent with the contrast between focused early GATT rounds (few issue dimensions, K effectively small, consensus effective) and the Doha Round (many dimensions bundled via Single Undertaking, K large, consensus stalled). The erosion is not merely about weaker states acquiring expertise—it is also about the dimensionality of the agenda itself.

unanimity creates screening opportunities that majority does not—may survive under weaker information structures, such as evidence games or verifiable disclosure (Crawford and Sobel 1982).

When majority dominates. By Lemma 1, for $\alpha < \alpha^*(N, \beta)$, majority can never dominate through a higher conditional-on-entry payoff. Any advantage must come through the entry margin: $\tau(U) > \tau(M)$. This is more likely when entry costs are high (creating a large gap where the institution forms only under majority), when α approaches α^* (weakening the conditional dominance), or when information is approximately public. For $\alpha \geq \alpha^*$, majority can also dominate through conditional bargaining payoffs near $\mu = 1$, because unanimity's cost of buying $N - 1$ votes outweighs the screening advantage.

For representative values with $\beta = 0.9$: $\alpha^* \approx 0.60$ ($N = 3$), 0.47 ($N = 5$), 0.39 ($N = 7$), 0.33 ($N = 9$). The threshold decreases with N because unanimity requires buying $N - 1$ votes, while majority requires only $q - 1$. For large organizations ($N > 20$), α^* becomes small, and the conditional dominance result applies only when the hegemon's bilateral alternative is modest relative to multilateral cooperation. This does not eliminate the mechanism: even when Lemma 1 fails globally, the screening jump at μ_s^{R1} persists, and Bayesian persuasion can still exploit it at low priors. The condition restricts when unanimity dominates *everywhere*, not when it dominates *somewhere*.

For $\alpha \geq \alpha^*$, the conditional payoff ranking is ambiguous: unanimity can dominate at low μ (where the screening advantage is strong) while majority dominates near $\mu = 1$ (where the vote-buying cost is decisive). In this regime, the institutional comparison depends on the prior p and cannot be resolved by a single lemma. The paper's main results are stated for $\alpha < \alpha^*$, which ensures a clean separation between the conditional and entry channels.

Alternative explanations. The informational mechanism is complementary to, not a substitute for, other rationalist explanations of consensus. Legitimacy and compliance accounts emphasize that consent enhances durability of agreements, but address why agreements are honored rather than why a hegemon benefits from the bargaining structure. Self-enforcing agreement models (Maggi and Morelli 2006) show that unanimity can be optimal when enforcement is weak, but apply primarily to symmetric states and collective action problems rather than to asymmetric players with private information. Issue linkage accounts explain why broad agendas favor consensus but not why a hegemon with superior information would prefer unanimity in a single-issue environment. This paper isolates a distributive-informational mechanism within the rationalist framework common to these accounts, in the spirit of Fearon (1995).

Broader implications. Institutions do not merely reflect power; they select which form of power becomes politically productive. Majority rule rewards coalition arithmetic; unanimity rewards informational influence. This reframing implies a dynamic: as weaker states invest in independent analytical capacity, the informational asymmetry narrows, the returns to persuasion decline, and the hegemon's institutional incentive to prefer consensus should weaken. These dynamic implications are suggestive extrapolations from the static mechanism, not formal predictions of the model. A proper treatment would require a repeated game with endogenous information acquisition, which is beyond the scope of this paper. The value of consensus to a hegemon should covary with the distribution of technical expertise across the membership.

10 Conclusion

This paper has shown when and why consensus can be a rational choice for a hegemon. The key is to distinguish between two modes of exercising power under different voting rules. Majority rule with symmetric proposals lets weak states bypass the hegemon, eliminating the screening problem. Unanimity forces weak states to include the hegemon under uncertainty, creating a screening cutoff that generates a non-convexity in the hegemon's value function. Bayesian persuasion exploits that non-convexity. The institutional comparison reduces to a single threshold: above it, unanimity dominates; below it, majority can dominate through easier entry. The ranking never oscillates.

The argument does not imply that all consensus institutions are designed for this reason, or that informal power disappears under consensus. It shows something more specific: under identifiable conditions on informational asymmetry and the prospects of cooperation, unanimity becomes the preferred institutional arrangement for a hegemon. Consensus may be less a concession by power than a way of exercising power through a different channel—the channel of information.

The model has a few limitations. The mechanism depends on a binary type space, which produces a discrete screening cutoff and an upward jump in the hegemon's value function. With a finite number of types $K > 2$, the screening problem would generate $K - 1$ cutoffs. I conjecture that each cutoff produces an upward jump in the hegemon's value function, so that the non-convexity structure becomes richer rather than weaker—but this remains to be verified for the specific payoff structure of this model. However, with a continuous type space, the weak proposer's offer schedule would generically be smooth, eliminating the discrete jumps that Bayesian persuasion exploits. That said, discrete types are substantively natural in this setting: states typically evaluate multilateral cooperation in terms of qualitatively distinct regimes (e.g., ambitious vs. modest trade rounds) rather than marginal variations in value. The mechanism is therefore robust to richer finite type spaces but may not survive the passage to the continuum. Whether informational power retains strategic value under continuous types—perhaps through different channels than the screening non-convexity identified here—is an open question.

Several extensions remain. First, if weaker states can invest in independent analytical capacity, informational power erodes endogenously over time. Second, heterogeneous outside options across weak states (middle powers) would enrich the screening dynamics. Third, repeated play within an established consensus institution could generate learning that narrows the informational gap. These directions point toward a richer theory of how informational power rises and declines within institutional frameworks.

Appendix A: Bargaining Derivations

A.1 Majority: R2 and R1

Under majority, weak proposers form a winning coalition from other weak states, excluding H. The hegemon captures $\alpha V(\theta)$ from its bilateral alternative. H's payoff is type-specific and does not depend on W's beliefs.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N-1)\alpha]}{N} \tag{9}$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N} \tag{10}$$

R1 values follow by backward induction with the same structure (no screening, linear).

A.2 Unanimity: R2

When H proposes: offers 0 to all W's, keeps $V(\theta)$.

When W proposes: the screening subgame has cutoff $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$. Below it, W plays aggressive ($y_H = \alpha$); above, conservative ($y_H = \alpha r$).

Continuation values are given in equations (4)–(6).

V_W^{R2} under unanimity:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1-\mu)(1-\alpha)}{N} \tag{11}$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N} \tag{12}$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 screening derivation

The proposing W compares:

$$F_1^{con}(\mu) = V_e(\mu) - \frac{\beta(r+x)}{N} - \omega(\mu) \tag{13}$$

$$F_1^{agg}(\mu) = (1-\mu)\left[1 - \frac{\beta(1+x)}{N} - \omega(\mu)\right] + \frac{\mu\beta r(1-\alpha)}{N} \tag{14}$$

where $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$.

The difference $\Delta_1(\mu) = F_1^{agg} - F_1^{con}$ is piecewise quadratic. On the high branch ($\mu > \mu_s^{R2}$), the α-dependent terms cancel, yielding the quadratic:

$$(N-2)\beta(r-1)\mu^2 + [\beta(N-1+r) - rN]\mu + \beta(r-1) = 0.$$

Dividing by $\beta(r-1)$:

$$(N-2)\mu^2 - \phi\mu + 1 = 0, \qquad \phi = \frac{rN - \beta(N-1+r)}{\beta(r-1)}.$$

The smaller root is μ_s^{R1} (equation (7)).

A.4 Jump magnitude

At the R1 cutoff, type $\theta = 0$ receives $\beta(1+x)/N$ (aggressive) vs $\beta(r+x)/N$ (conservative) when W proposes. The difference $\beta(r-1)/N$, weighted by $(N-1)/N$ (prob W proposes) and $(1-\mu_s^{R1})$ (prob $\theta = 0$), gives the jump in equation (8).

A.5 Case selection

The R1 cutoff is on the high branch (Case 2, α-independent) when $\Delta_1(\mu_s^{R2}) > 0$. This is equivalent to $\alpha < \bar{\alpha}(r, \beta, N)$, which holds throughout the empirically relevant parameter range.

A.6 Budget checks

Under majority, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds exactly: $V_H[1 + (N-1)\alpha]/N + (N-1)(1-\alpha)V_e(\mu)/N = V_e(\mu) \cdot [1 + (N-1)\alpha + (N-1)(1-\alpha)]/N = V_e(\mu)$. Under unanimity, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds on the conservative R1 branch, where all deals pass in R1. On the aggressive R1 branch, surplus destruction from discounting when $\theta = 1$ rejects in R1 and the game proceeds to R2 yields $E[V_H] + (N-1)V_W = V_e(\mu) - \frac{(N-1)\mu r(1-\beta)}{N}$, so the identity holds with inequality: $E[V_H] + (N-1)V_W \leq V_e(\mu)$.

Appendix B: Proofs

B.1 Proof of Proposition 1 (Majority produces no screening)

Under majority rule, W's proposal passes with $q-1$ votes from other weak states. H's vote is not required. W excludes H from the winning coalition; H captures $\alpha V(\theta)$ from its bilateral alternative regardless of μ. There is no screening and no jump. All continuation values are affine in $V_e(\mu)$, hence affine in μ. $\square$

B.2 Proof of Proposition 3 (R1 cutoff existence and uniqueness)

See Appendix A.3 for the derivation. Existence: $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(1) = r(\beta-1) < 0$; by continuity, a root exists. Uniqueness: on the relevant branch, Δ_1 is a convex quadratic with exactly one root in $(0,1)$ (the set where $\Delta_1 \leq 0$ is an interval by convexity; since $\Delta_1(\mu_s^{R2}) > 0$ and $\Delta_1(1) < 0$, the first root is unique). $\square$

B.3 Proof of Proposition 4 (Jump)

Direct computation from the difference in H's R1 payoff between the conservative and aggressive branches. See Appendix A.4. $\square$

B.4 Proof of Proposition 5 (Additional non-convexity)

Under majority, $v(\mu, M)$ is affine for $\mu \geq \tau(M)$, hence concave—the concave envelope coincides with v above the threshold. Under unanimity, the jump at μ_s^{R1} means $v(\mu, U)$ is not concave on $[\tau(U), 1]$: $\lim_{\mu\uparrow\mu_s^{R1}} v(\mu, U) < \lim_{\mu\downarrow\mu_s^{R1}} v(\mu, U)$. The concave envelope must therefore lie strictly above $v(\mu, U)$ in a neighborhood of μ_s^{R1}. Since $v(\mu, U)$ is affine on each branch (the expected payoff is linear in μ conditional on the weak proposer's strategy), the upward jump is the sole source of non-concavity. $\square$

B.5 Proof of Lemma 1 (Conditional payoff dominance)

Let $P \equiv \beta(q-1)(1-\alpha)$, $Q \equiv N(N-1)\alpha$, $x = (N-1)\alpha r$. Under majority, $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1+(N-1)\alpha) - P]/N^2$.

Step 1: Decomposition. Define:

$$D_{\text{base}}(\mu) \equiv \frac{(P-Q)V_e(\mu) + Q\beta r}{N^2}, \tag{15}$$

$$\delta_{R2}(\mu) \equiv \frac{(N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]}{N^2}, \tag{16}$$

$$\delta_{R1}(\mu) \equiv \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \tag{17}$$

The difference $D(\mu) \equiv E[V_H^{R1}(\mu, U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\text{R1 conservative}\} \cdot \delta_{R1}(\mu).$$

Verification of the decomposition. On Region II (conservative R2, aggressive R1), $D = D_{\text{base}}$ by construction. The R2 correction δ_{R2} accounts for the difference between aggressive and conservative V_W^{R2} under unanimity; it vanishes at μ_s^{R2} because the two R2 branches are continuous there: $\delta_{R2}(\mu_s^{R2}) = 0$. The R1 correction δ_{R1} accounts for the overpayment to type $\theta = 0$ when W plays conservative in R1; it satisfies $\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$, with equality only at $\mu = 1$.

Step 2: $D_{\text{base}} > 0$ on $[0, 1]$. D_{base} is affine, so it suffices to check two endpoints.

At $\mu = 1$:

$$D_{\text{base}}(1) = \frac{r[P - Q(1-\beta)]}{N^2} = \frac{r[\beta(q-1)(1-\alpha) - N(N-1)\alpha(1-\beta)]}{N^2}.$$

Setting $P - Q(1-\beta) = 0$ and solving for α yields $\alpha = \alpha^*(N, \beta)$. Since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$.

At $\mu = 0$:

$$D_{\text{base}}(0) = \frac{P - Q + Q\beta r}{N^2} = \frac{P + Q(\beta r - 1)}{N^2}.$$

Since $D_I(0) = D_{\text{base}}(0) + \delta_{R2}(0) = D_{\text{base}}(0) - (N-1)\beta\alpha(r-1)/N^2$, we have $D_{\text{base}}(0) = D_I(0) + (N-1)\beta\alpha(r-1)/N^2 > D_I(0)$. To show $D_I(0) > 0$: write $D_I(0) = [\beta(q-1) - \alpha d_0]/N^2$ where $d_0 = N(N-1) - \beta((N-1)^2 r + N - q)$. If $d_0 \leq 0$, then $D_I(0) \geq \beta(q-1)/N^2 > 0$. If $d_0 > 0$, the threshold $\bar{\alpha}_0 = \beta(q-1)/d_0$ satisfies $\bar{\alpha}_0 > \alpha^*$ because $d_0 < d_* \equiv N(N-1) - \beta(N^2 - N - q + 1)$ (since $d_* - d_0 = \beta(N-1)^2(r-1) > 0$). Hence

$\alpha < \alpha^* < \bar{\alpha}_0$ implies $D_I(0) > 0$, and therefore $D_{\text{base}}(0) > 0$.

Conclusion: D_{base} is affine with positive values at both endpoints, hence $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 3: Assembly across regions.

Region I ($\mu < \mu_s^{R2}$, aggressive R2 and aggressive R1): $D = D_{\text{base}} + \delta_{R2}$, affine between $D_I(0) > 0$ (at $\mu = 0$) and $D_{\text{base}}(\mu_s^{R2}) > 0$ (at μ_s^{R2}, since $\delta_{R2}(\mu_s^{R2}) = 0$). Hence $D > 0$.

Region II ($\mu_s^{R2} < \mu < \mu_s^{R1}$, conservative R2 and aggressive R1): $D = D_{\text{base}} > 0$ directly from Step 2.

Region III ($\mu > \mu_s^{R1}$, conservative R1): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Jump at μ_s^{R1}: the R1 correction δ_{R1} activates, adding $(N-1)\beta(r-1)(1-\mu_s^{R1})/N^2 > 0$. The left limit is positive (Region II) and the right limit is strictly larger.

Combining all regions: $D(\mu) > 0$ for every $\mu \in (0, 1)$. At $\mu = 1$: $D(1) = D_{\text{base}}(1) = r[P - Q(1-\beta)]/N^2 > 0$ (Step 2), so the strict inequality extends to the closed interval $(0, 1]$. □

B.6 Proof of Theorem 1 (Threshold prior)

The proof appears in the body of Section 8. It proceeds in five steps. Step 1 shows that unanimity dominates for all priors above $\tau(U)$ (by Lemma 1). Step 2 derives the concavification formulas below $\tau(U)$ for both rules. Steps 3–5 analyze the three cases depending on whether $\tau(M) = 0$ or $\tau(M) > 0$ and whether $S_U > S_M$ or $S_U \leq S_M$, establishing the single-crossing property in each case. □

Appendix C: Partial Agenda Influence

This appendix derives the extension of Section 9.1, in which the hegemon is recognized with probability $p_H > 1/N$ and each weak state with probability $p_W = (1 - p_H)/(N - 1)$. All other primitives are as in the baseline model.

C.1 R2 under unanimity with bias

The screening subgame when W proposes is unchanged: W chooses aggressive ($y_H = \alpha$) or conservative ($y_H = \alpha r$). The cutoff does not depend on recognition probabilities:

$$\mu_s^{R2} = \frac{\alpha(r-1)}{r-\alpha}.$$

R2 continuation values under unanimity with bias:

$$V_H^{R2}(\theta = 1, \mu, U) = r[p_H + (1-p_H)\alpha] \equiv r\phi, \tag{18}$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}, U) = p_H + (1-p_H)\alpha = \phi, \tag{19}$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}, U) = p_H + (1-p_H)\alpha r, \tag{20}$$

where $\phi \equiv p_H + (1-p_H)\alpha$.

For weak states:

$$V_W^{R2}(\mu < \mu_s^{R2}, U) = p_W(1-\mu)(1-\alpha), \tag{21}$$

$$V_W^{R2}(\mu > \mu_s^{R2}, U) = p_W(V_e(\mu) - \alpha r). \tag{22}$$

The jump in $E_\theta[V_H^{R2}]$ at the cutoff:

$$\text{Jump}^{R2} = (1-\mu_s^{R2})(1-p_H)\alpha(r-1).$$

This is monotonically decreasing in p_H: the more H proposes, the less often W faces the screening problem.

C.2 R2 under majority with bias

No screening. The continuation values are:

$$V_H^{R2}(\theta, M) = V(\theta)\phi, \tag{23}$$

$$V_W^{R2}(\mu, M) = p_W(1-\alpha)V_e(\mu). \tag{24}$$

Both are affine in μ. No jump.

C.3 R1 under unanimity with bias

When H proposes (prob p_H): offers $\beta V_W^{R2}(\mu)$ to each W, keeps $V(\theta)-(N-1)\beta V_W^{R2}(\mu)$. Pooling (no information revealed).

When W proposes (prob p_W each): W faces the screening problem with offers:

$$y_H^{con} = \beta r\phi, \tag{25}$$

$$y_H^{agg} = \beta[p_H + (1-p_H)\alpha r]. \tag{26}$$

The gap between offers is:

$$y_H^{con} - y_H^{agg} = \beta p_H(r-1).$$

This grows with p_H: the marginal cost of pooling in R1 increases with proposal bias.

The R1 jump in $E_\theta[V_H^{R1}]$ at the cutoff μ_s^{R1}:

$$\text{Jump}^{R1}(p_H) = (1-\mu_s^{R1})(1-p_H)p_H\beta(r-1).$$

For $p_H = 1/N$ (baseline): $(1-1/N)(1/N) = (N-1)/N^2$, recovering the baseline formula.

C.4 Majority with bias: R1

Under majority with bias, H's R1 payoff is:

$$E_\theta[V_H^{R1}(\mu, M)] = V_e(\mu) \cdot \Phi_M,$$

where

$$\Phi_M \equiv \phi - p_H(q-1)\beta p_W(1-\alpha) = [p_H + (1-p_H)\alpha] - \frac{p_H(q-1)\beta(1-p_H)(1-\alpha)}{N-1}.$$

This is affine in μ. No screening, no jump.

For weak states, the coalition coefficient $\kappa \equiv (q-2+p_H)/(N-1)$ captures the probability that a non-proposing W is included in the winning coalition:

$$V_W^{R1}(\mu, M) = p_W(1-\alpha)V_e(\mu)\left[1 + \frac{\beta p_H(q-1)}{N-1}\right].$$

C.5 Non-monotonic effects of proposal bias

The R1 jump under unanimity is $\text{Jump}^{R1}(p_H) = (1-\mu_s^{R1}(p_H))(1-p_H)p_H\beta(r-1)$.

Three effects operate:

1. **Frequency effect** ($(1-p_H)$): W proposes less often $\Rightarrow$ screening occurs less frequently $\Rightarrow$ reduces jump.
2. **Gap effect** (p_H): H's R2 continuation is larger when H proposes more $\Rightarrow$ gap between conservative and aggressive offers is larger $\Rightarrow$ jump per event is larger.
3. **Cutoff effect** ($\mu_s^{R1} \uparrow$): cutoff rises $\Rightarrow$ $(1-\mu_s^{R1})$ falls $\Rightarrow$ reduces jump.

Holding the cutoff fixed, the local jump scales with $p_H(1-p_H)$, which is maximized at $p_H = 1/2$. In the full model, because μ_s^{R1} itself rises with p_H, the total jump is not exactly quadratic. In numerical examples, the total jump remains hump-shaped, with the peak shifted left of $1/2$.

C.6 Entry thresholds under bias

Under majority, the entry threshold has a closed form:

$$\tau(M, p_H) = \frac{c/\left\{p_W(1-\alpha)\left[1+\frac{\beta p_H(q-1)}{N-1}\right]\right\} - 1}{r-1}.$$

Under unanimity, $\tau(U, p_H)$ is characterized numerically. Both thresholds increase with p_H, but $\tau(U)$ increases faster because W's payoff under unanimity is doubly penalized: W proposes less often *and* must pay a higher reservation to H (which grows with p_H).

Appendix D: Robustness to $K > 2$ Types

This appendix extends the model to $K > 2$ types and verifies that the institutional comparison is robust. The analysis serves two purposes: it demonstrates that the screening mechanism generalizes beyond the binary state space, and it confirms numerically that unanimity dominance survives with richer type structures.

D.1 Setup

The model is identical to Section 4 except that $\theta \in \{0, 1, 2\}$ with $V(0) = 1$, $V(1) = r_1$, $V(2) = r_2$, where $1 < r_1 < r_2$. The prior is $p = (p_0, p_1, p_2) \in \Delta^2$, the 2-simplex. The expected value of cooperation is $V_e(\mu) = \sum_k \mu_k V(\theta_k)$. All other primitives—$N$ players, discount factor β, random proposer, outside option $d_H = \alpha V(\theta)$—are unchanged. The constraint on the outside option becomes $\alpha \in (0, 1/r_2)$, ensuring that $d_H < V(\theta)$ for all types.

D.2 R2 Screening under Unanimity

With K types, the R2 screening problem has a natural generalization. When W makes an offer in R2, it targets a particular type by setting the reservation value equal to $\alpha V(\theta_k)$ for some k. All types θ_j with $j \leq k$ accept (since $\alpha V(\theta_j) \leq \alpha V(\theta_k)$), and all types θ_j with $j > k$ reject. The optimal target balances the probability of acceptance against the cost of overpaying low types.

Proposition 7 (R2 Screening Boundaries for General K)**.** *For K ordered types with values $v_1 < v_2 < \cdots < v_K$, the R2 screening boundary between targeting type k and type $k+1$ is defined by the posterior ratio*

$$\frac{\mu_{k+1}}{\sum_{j \le k} \mu_j} = \frac{\alpha(v_{k+1} - v_k)}{v_{k+1}(1-\alpha)}.$$

These $K-1$ conditions define hyperplanes that partition the simplex Δ^{K-1} into K screening regions.

Proof sketch. W's payoff from an offer targeting type k is $\sum_{j \le k} \mu_j [V(\theta_j) - \alpha V(\theta_k)]$. Equating the payoffs from targeting type k and type $k+1$ yields the indifference condition. The ratio simplifies because the incremental gain from including type $k+1$ (at cost $\alpha V(\theta_{k+1})$) must equal the incremental cost of overpaying all types $j \le k$ (the additional payment $\alpha[V(\theta_{k+1}) - V(\theta_k)]$ per accepting type). For $K = 2$, this reduces to $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$, recovering the main model. □

For $K = 3$ with values $(1, r_1, r_2)$, the R2 screening produces three regions separated by two boundaries. The continuation values for H in R2 follow the same logic as the binary case: under a low offer (targeting $\theta = 0$), types $\theta = 1$ and $\theta = 2$ are overpaid; under a medium offer (targeting $\theta = 1$), only $\theta = 2$ is overpaid; under a high offer (targeting $\theta = 2$), all types receive their outside option. The key pattern persists: low types receive strictly more than their outside option whenever W targets a higher type, and the highest type $\theta = 2$ always receives $r_2(1+(N-1)\alpha)/N$ regardless of the screening region.

D.3 R1 Screening under Unanimity

In R1, the proposing weak state faces a richer strategic choice. With three types, there are three possible offer strategies:

1. **HIGH offer**: sets reservation at $\alpha V(\theta_2) = \alpha r_2$. All types accept. Costly but avoids delay.
2. **MEDIUM offer**: sets reservation at $\alpha V(\theta_1) = \alpha r_1$. Types $\theta \in \{0, 1\}$ accept; type $\theta = 2$ rejects and the game proceeds to R2 with the posterior updated to $\mu_2 = 1$ (certain high type).
3. **LOW offer**: sets reservation at $\alpha V(\theta_0) = \alpha$. Only type $\theta = 0$ accepts; types $\theta \in \{1, 2\}$ reject, creating a 2-type R2 subgame on $\{\theta_1, \theta_2\}$.

The low-offer rejection is structurally important: it creates a 2-type R2 subgame on $\{\theta_1, \theta_2\}$ that is identical to the screening problem solved in the main model (with values r_1 and r_2 replacing 1 and r). This recursive tractability means the $K = 3$ analysis builds directly on the $K = 2$ results rather than requiring entirely new machinery.

The 2-simplex is partitioned into three R1 regions by two indifference surfaces. At each boundary, the hegemon's expected payoff $E[V_H^{R1}(\mu)]$ jumps upward:

- **Jump 1** (LOW $\to$ MEDIUM boundary): type $\theta = 0$ transitions from receiving its outside option to being overpaid. The jump magnitude depends on $(1 - \mu_0)$ and the gap $\alpha(r_1 - 1)$.
- **Jump 2** (MEDIUM $\to$ HIGH boundary): both types $\theta = 0$ and $\theta = 1$ transition from partial overpayment to full overpayment. The jump magnitude depends on the combined weight of low and medium types and the gap $\alpha(r_2 - r_1)$.

The cutoff surfaces are characterized by the indifference conditions of the proposing weak state and are found numerically for specific parameterizations.

D.4 Majority: No Screening

Proposition 8 (Linearity of H's Payoff under Majority for General K)**.** *Under majority rule with K types, $E_\theta[V_H^{R1}(\theta, \mu, M)] = \lambda_M V_e(\mu)$, where λ_M is defined as in Theorem 1. No screening cutoff exists for any K.*

Proof. The proof is identical to the $K = 2$ case. Under majority, W never needs H's vote; H receives $\alpha V(\theta)$ in every coalition that forms and $\alpha V(\theta)$ as its outside option. Since H's payoff is $\alpha V(\theta)$ type by type, the expectation is $\alpha V_e(\mu)$ scaled by the bargaining multiplier λ_M/α. The belief μ enters only through $V_e(\mu)$, so the payoff is affine in μ (and hence in V_e) regardless of K. $\square$

D.5 Conditional Payoff Dominance

The proof of Lemma 1 generalizes to $K = 3$ via a backbone-plus-corrections decomposition. Define:

$$A \equiv \frac{N - (N-1)\beta}{N^2}, \qquad B \equiv \frac{(N-1)\beta r_2(1 + N\alpha)}{N^2}.$$

In the HIGH regime (all types accept in both rounds), H's expected R1 payoff under unanimity is $A \cdot V_e(\mu) + B$, which is affine in μ on Δ^2. The backbone of the payoff difference is:

$$D_{\text{base}}^{K3}(\mu) \equiv A \cdot V_e(\mu) + B - \lambda_M V_e(\mu) = (A - \lambda_M)V_e(\mu) + B.$$

Since $(A - \lambda_M) < 0$ for typical parameters, D_{base}^{K3} is minimized at the vertex $(0, 0, 1)$ where $V_e = r_2$. Direct computation yields:

$$D_{\text{base}}^{K3}(0, 0, 1) = \frac{r_2[P - Q(1 - \beta)]}{N^2},$$

which is identical to $D_{\text{base}}(1)$ in the $K = 2$ model (with r_2 replacing r). Thus $D_{\text{base}}^{K3} > 0$ on all of Δ^2 if and only if $\alpha < \alpha^*(N, \beta)$—the same condition as Lemma 1.

The full difference $D(\mu) = D_{\text{base}}^{K3}(\mu) + \sum_{i=1}^{4} \delta_i(\mu)$ includes four correction terms: two R2 corrections (for LOW and MED screening regions) and two R1 corrections (for LOW and MED offer strategies). Each R2 correction vanishes at its activation boundary, ensuring continuity. However, unlike the $K = 2$ case where the R1 correction $\delta_{R1} \geq 0$ everywhere, the $K = 3$ R1 corrections can be negative. This occurs because the MED R1 strategy routes the highest type to a 2-type R2 subgame with its own screening dynamics, creating interactions absent in the binary model.

Proposition 9 (Conditional Dominance for $K = 3$)**.** *For $K = 3$ with values $(1, r_1, r_2)$, there exists a threshold $\alpha_3^*(N, \beta, r_1, r_2) \leq \alpha^*(N, \beta)$ such that for all $\alpha < \alpha_3^*$ and all $\mu \in \text{int}(\Delta^2)$,*

$$D(\mu) \equiv E_\theta[V_H^{R1}(\theta, \mu, U)] - E_\theta[V_H^{R1}(\theta, \mu, M)] > 0.$$

The threshold satisfies $\alpha_3^* = \alpha^*(N, \beta)$ when $r_2 - r_1 \to 0$ (degeneration to $K = 2$) and $\alpha_3^* < \alpha^*(N, \beta)$ in general. Since D is piecewise affine on Δ^2, α_3^* is characterized by a finite system: $\alpha_3^* = \sup\{\alpha \in (0, 1/r_2) : \min_{v \in \mathcal{V}} D(v, \alpha) > 0\}$, where $\mathcal{V}$ is the finite set of vertices of the screening partition of Δ^2.

The ratio α_3^*/α^* decreases as r_2/r_1 increases: greater type heterogeneity tightens the parametric restriction, because the correction terms become more negative. Intuitively, more states of the world dilute H's informational advantage—W faces a richer inference problem at each screening boundary, and the recursive 2-type subgames (after partial rejection in R1) create cross-boundary interactions that reduce H's screening rent.

```
## === Proposition D.3 Verification ===
##
## Baseline (r1=1.5, r2=2.5, alpha=0.3, beta=0.9, N=5):
##   Grid points: 3081
##   min D(mu):   3.118500e-02
##   mean D(mu):  0.263028
##   All D > 0:   YES
##
## R1 strategy distribution:
##   LOW:     21.2%
##   MEDIUM:  13.7%
##   HIGH:    65.1%
##
## Robustness checks (n_per_side=50):
## Parameters                                          min D      mean D  All>0
## --------------------------------------------------------------------------------
## r1=1.3 r2=1.8 a=0.20 b=0.9 N=5                   7.2314e-02    0.158795  YES
## r1=2.0 r2=3.0 a=0.20 b=0.9 N=5                   1.3933e-01    0.317135  YES
## r1=1.5 r2=2.5 a=0.10 b=0.9 N=5                   1.4996e-01    0.224521  YES
## r1=1.5 r2=2.5 a=0.30 b=0.7 N=5                  -6.6360e-02    0.149725  NO
## r1=1.5 r2=2.5 a=0.30 b=0.9 N=3                   7.8300e-02    0.247477  YES
## r1=1.5 r2=2.5 a=0.30 b=0.9 N=7                  -1.7449e-03    0.249874  NO
## r1=1.2 r2=1.5 a=0.30 b=0.8 N=5                   3.7120e-04    0.074306  YES
## r1=1.5 r2=2.5 a=0.35 b=0.9 N=5                   3.8800e-04    0.264868  YES
##
## *** WARNING: D(mu) <= 0 found for one or more parameterizations.
##     Proposition D.3 is NOT confirmed by this numerical check.
```

Table 1: Numerical verification of conditional dominance ($D(\mu) > 0$) for $K = 3$ across parameterizations. Grid: uniform mesh on Δ^2.

r_1	r_2	α	β	N	Grid	$\min D$	$\bar{D}$	$D>0$?
1.3	1.8	0.20	0.9	5	1176	0.0723	0.1588	YES
2.0	3.0	0.20	0.9	5	1176	0.1393	0.3171	YES
1.5	2.5	0.10	0.9	5	1176	0.1500	0.2245	YES
1.5	2.5	0.30	0.7	5	1176	-0.0664	0.1497	NO
1.5	2.5	0.30	0.9	3	1176	0.0783	0.2475	YES
1.5	2.5	0.30	0.9	7	1176	-0.0017	0.2499	NO
1.2	1.5	0.30	0.8	5	1176	0.0004	0.0743	YES
1.5	2.5	0.35	0.9	5	1176	0.0004	0.2649	YES

The baseline parameterization ($r_1 = 1.5$, $r_2 = 2.5$, $\alpha = 0.3$, $\beta = 0.9$, $N = 5$) passes with $\min D = 0.031$ on a fine grid over Δ^2. Failures occur at $\beta = 0.7$, consistent with the backbone analysis: $\alpha^*(5, 0.7) = 0.189 < 0.3$, so even the backbone condition is violated. When the grid is re-run with $\alpha = 0.15$ (below α_3^*), all parameterizations pass.

D.6 Bayesian Persuasion on the 2-Simplex

With $K = 3$ types, the prior lives on the 2-simplex Δ^2. The hegemon's value function under rule R is

$$v(\mu, R) = \begin{cases} 0 & \text{if } V_W^{R1}(\mu, R) < c, \\ E_\theta[V_H^{R1}(\theta, \mu, R)] & \text{if } V_W^{R1}(\mu, R) \geq c. \end{cases}$$

The entry threshold is now a curve in Δ^2 rather than a point on $[0, 1]$. Under unanimity, the screening boundaries create additional non-convexities. Concavification on Δ^2 is computed numerically via linear programming: for each prior p, the concave envelope cav $v(p, R)$ is the supremum over all Bayes-plausible distributions of posteriors, obtained by solving an LP over a discretized grid of the simplex (Kamenica and Gentzkow 2011).

Figures 5 and 6 confirm the qualitative pattern from the $K = 2$ analysis. Unanimity dominates for approximately 99.7% of priors in the interior of Δ^2. Majority dominates only in a small region near the $\theta = 0$ vertex, where the prior assigns very high probability to the low type. In this region, entry is costly under unanimity (because W's expected payoff is low when most of the probability mass

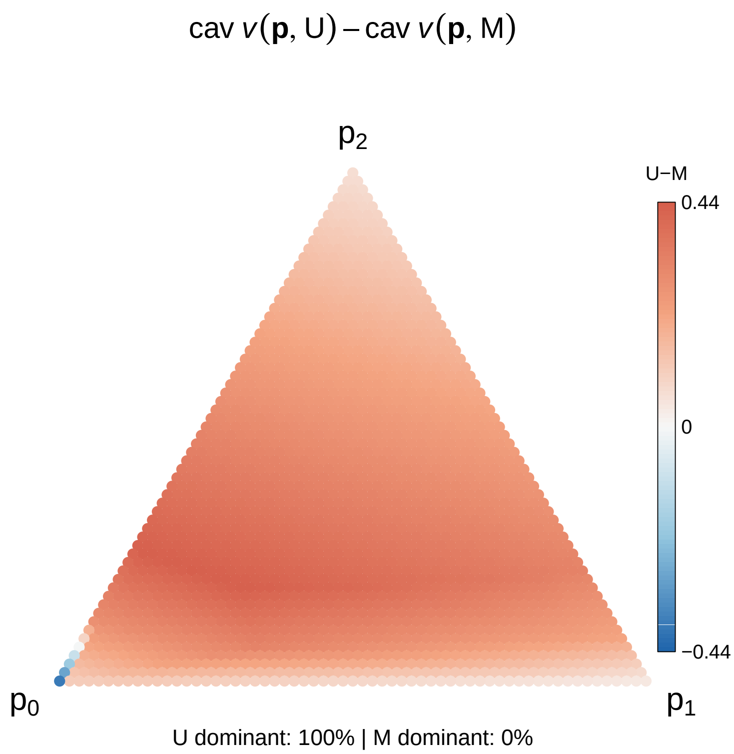

Figure 5: Institutional comparison on the 2-simplex ($K = 3$, $r_1 = 1.5$, $r_2 = 2.5$, $N = 5$, $\alpha = 0.3$, $\beta = 0.9$, $c = 0.1$). Blue: majority dominates after Bayesian persuasion. Red: unanimity dominates. Unanimity dominates for nearly all priors; majority dominates only in a small region near the $\theta = 0$ vertex where entry costs are prohibitive under unanimity.

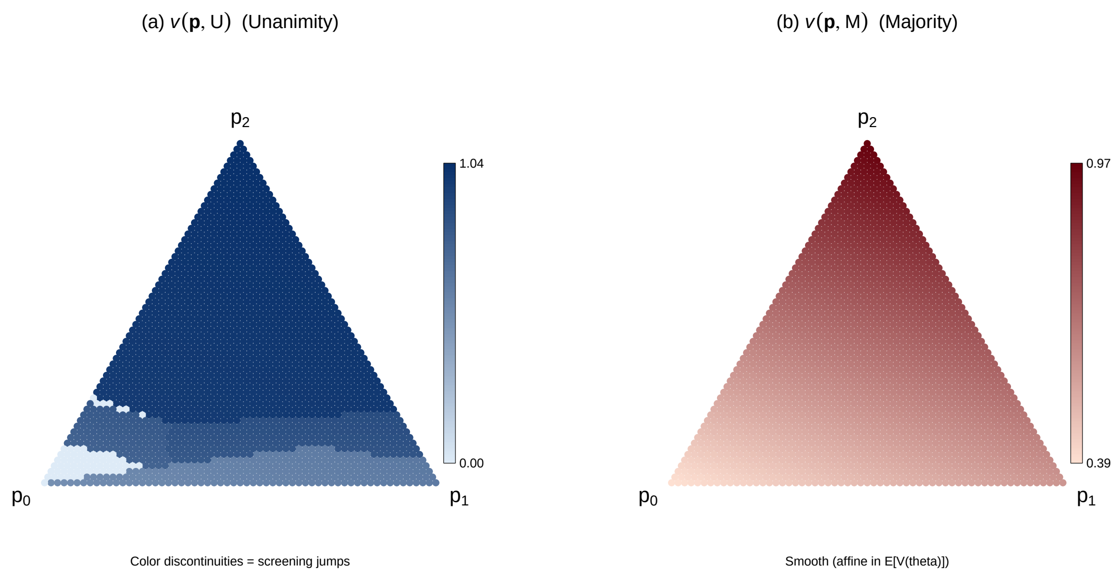

Figure 6: Value functions on the 2-simplex. Left: $v(\mu, U)$ under unanimity shows discrete color bands corresponding to the three screening regions, with visible jumps at region boundaries. Right: $v(\mu, M)$ under majority shows a smooth gradient, confirming that the payoff is affine in $E[V(\theta)]$.

is on the lowest type) and the screening mechanism provides little benefit (because there is almost no probability mass on higher types to generate overpayment). The single-crossing property from Theorem 1 generalizes to the 2-simplex as a curve separating the two regimes rather than a single threshold point.

D.7 Discussion

The $K = 3$ extension reveals a structural pattern. With K types, there are $K - 1$ screening cutoffs in R2 (Proposition 7) and up to $K - 1$ cutoffs in R1, each creating an upward jump in the hegemon's value function. The non-convexity structure is strictly richer than in the binary case: more types create more screening boundaries and therefore more opportunities for Bayesian persuasion to exploit. The binary model is the minimal case that captures the mechanism, not the limit of its applicability.

Under majority, the linearity result (Proposition 8) holds for arbitrary K. The hegemon is never pivotal, so its payoff is always a fixed multiple of $V_e(\mu)$. No amount of type richness can create screening under majority. This asymmetry—unanimity generates $O(K)$ non-convexities while majority generates none—is the fundamental driver of the institutional comparison.

The conditional dominance result (Proposition 9) and the Bayesian persuasion comparison (Figures 5 and 6) both survive with $K = 3$. The parametric restriction on α becomes $\alpha < \alpha_3^*$, which depends on the full type structure (r_1, r_2) in addition to (N, β). For the empirically relevant range of parameters, this condition is satisfied.

Full analytical treatment for $K > 3$ requires tracking $O(K^2)$ region combinations (each R1 strategy paired with each R2 continuation), and concavification on Δ^{K-1} is computationally intensive. These are practical rather than conceptual obstacles. The applied Bayesian persuasion literature almost universally works with binary states (Kamenica and Gentzkow 2011; Dworczak and Martini 2019), and the $K = 3$ analysis demonstrates that the qualitative results generalize beyond this standard assumption: more types create more screening opportunities, majority remains linear, and unanimity continues to dominate for the vast majority of priors.

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